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Saveetha Institute of Medical And Technical Sciences

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BTA0166 - BIOLOGY ENVIRONMENTAL SCIENCE

BHADLA AND PAVAGADA SOLAR PARKS - INTEGRATING AGRIVOLTAICS AND LAND-USE INNOVATION

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INTRODUCTION

- India leads in solar energy expansion .
- Combine solar energy and agriculture for sustainable development
- Bhadla (Rajasthan) and Pavagada (Karnataka) are two of the world’s largest solar parks.
- This study explores how solar energy and agriculture can co-exist for sustainable growth

Objectives

- Agrivoltaic -“Agri” (farming) + “Voltaic” (solar energy).
- Land used for dual purpose: farming + solar energy.
- Crops grown under or around solar panels.
- Improves land efficiency and income.

Benefits:

Dual income: crops + energy

Reduces heat and water loss

Improves panel performance

Supports rural economy



Literature Review

- Both **Bhadla (Rajasthan) and Pavagada (Karnataka)** are among the world's largest solar parks - over **2 GW each.**
- Studies highlight India's success in **large-scale solar deployment.**
- Literature focuses on **technical performance, land-use management, and socio-economic impacts**.
- New research emphasizes **agrivoltaics** – combining farming with solar installatioty.
- Agrivoltaics proposed to balance **energy generation and agriculture on same land.**

System Design / Architecture

1. <u>Bhadla Solar Park</u>

(Rajasthan)

Type: Utility-scale solar park

Capacity: ~2,245 MW (one of the largest in India)

Area: ~14,000 acres



2. <u>Pavagada Solar Park</u>

(Rajasthan)

Type: Utility-scale solar park community involvement

Capacity: ~2,050 MW

Area: ~13,000 acres



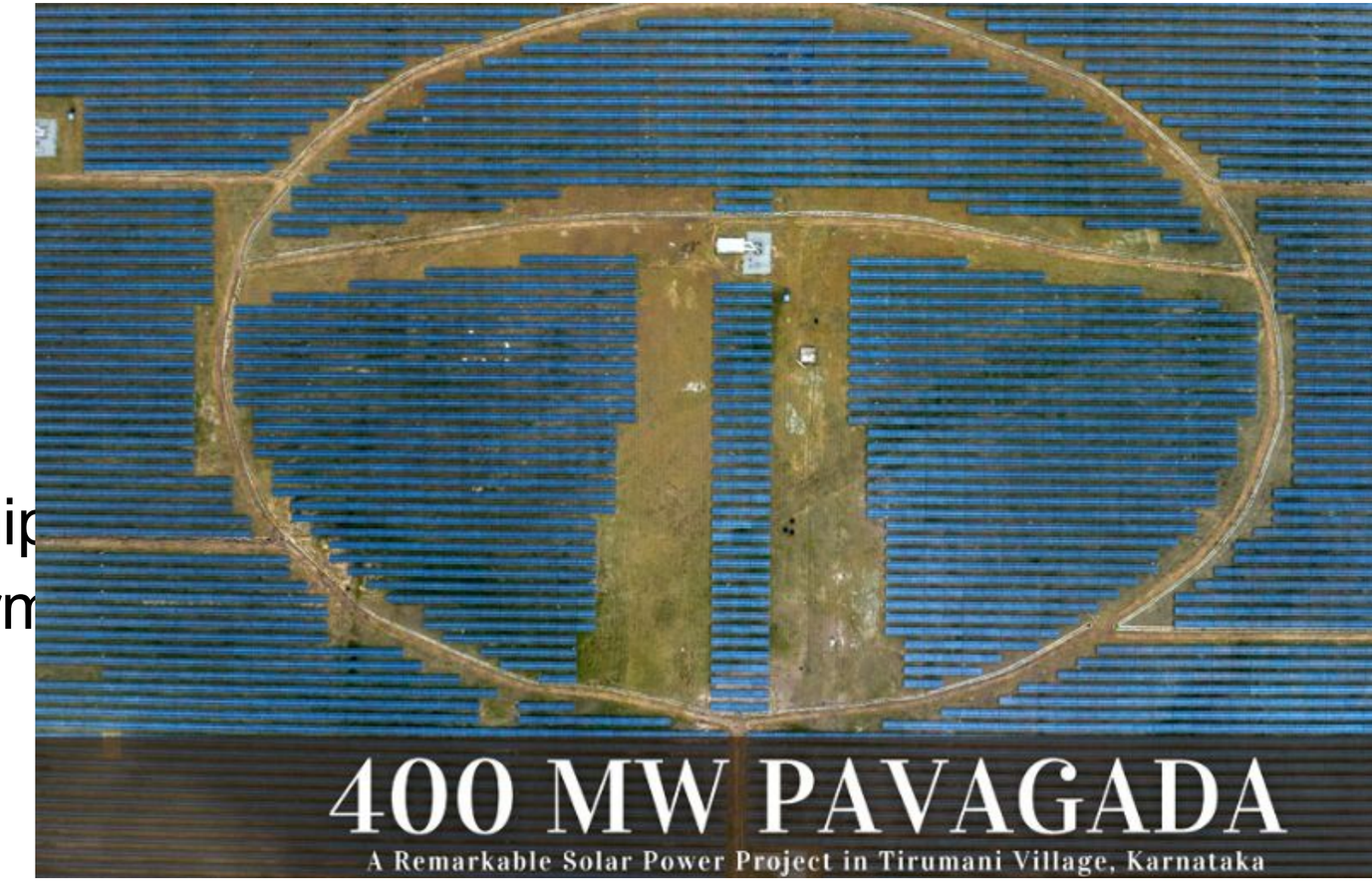
Bhadla Solar Park

Location : Rajasthan
Capacity:~2,245 MW
Area:14,000 acres
Features: High solar radiation
Potential:Grow dryland crops like (millets, pulses) under panels
Goal: Combine renewable power with local farming resilience



Pavagada Solar Park

Location : Karnataka
Capacity:~2,050 MW
Area:13,000 acres
Unique model: Land leased from farmers
Potential:Farmers receive annual rent.
Benefits: Local employment and ownership
Opportunity for agrivoltaic farm
- groundnuts, pulses, fodder



Challenges in Large Solar Parks

- High land requirement
- Biodiversity and soil degradation
- Limited community benefits
- Heat and dust affect crops nearby

Solution:

Agrivoltaic integration for balanced development



Implementation

Economic:
Dual income (crop + solar lease)
Reduces migration
Environmental:
Soil protection and moisture retention
Biodiversity support
Energy:
Increased efficiency due to panel cooling
Social:
Community inclusion
Rural job creation



Challenges

Co-use land model:
Solar + crops + grazing

Smart Farming:
AI & sensors for irrigation

Pollinator plants:
Boost Ecosystem

Land-leasing model:
Farmer participation



Future Vision

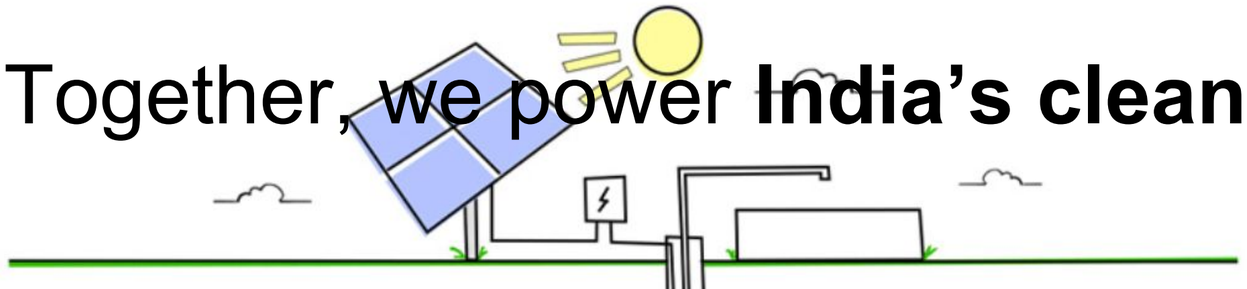
- India → global leader in agrivoltaic innovation
- Solar parks become “energy-farming hubs”
- Create Green Energy Villages
- Align with Net Zero 2070 target
- Promote inclusive and sustainable rural growth



CONCLUSION

- **Agrivoltaics** = Balance between energy, land, and people.
- **Bhadla & Pavagada** → global models of green innovation.
- Sustainable path for **energy** and **agriculture.**
- Together, we power **India’s clean future .**





References

1. **Bhadla Solar Park (Rajasthan)** - project summary, development in phases, capacity, land area.
2. **Ministry of New & Renewable Energy, Government of India**- Agrivoltaics / New & Innovative Solar Applications: Policy and regulatory framework.
3. **Karnataka, Government of India, District Tumakuru district planning document)- PLP 2025-26:** Rajasthan Renewable Energy Corporation Ltd. (RRECL)
4. **Implementation and developer data for Bhadla Solar Park Renewable Energy** – Pavagada Solar Park details(June 2023).


THANK YOU